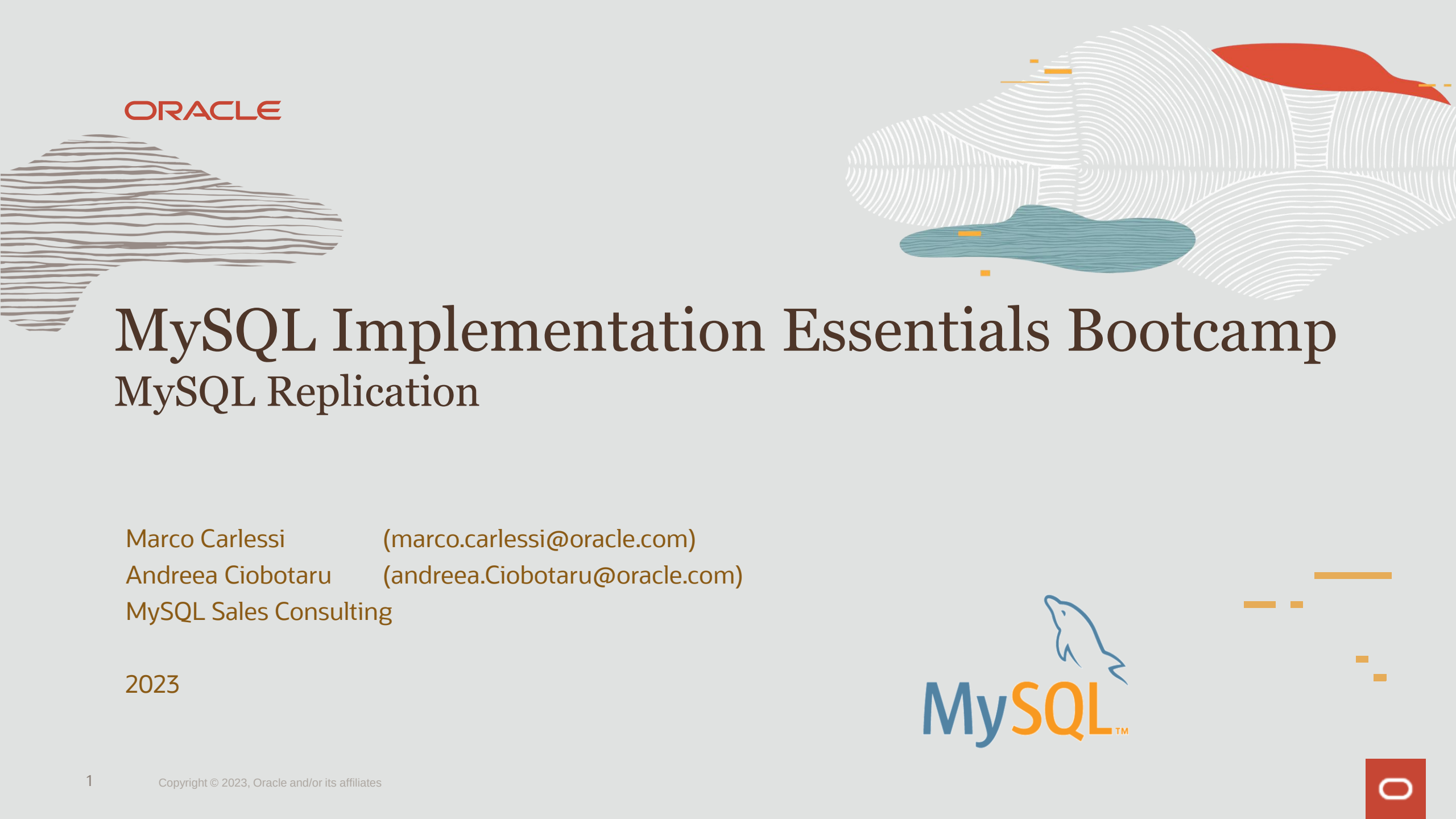




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MySQL Implementation Essentials Bootcamp

MySQL Replication

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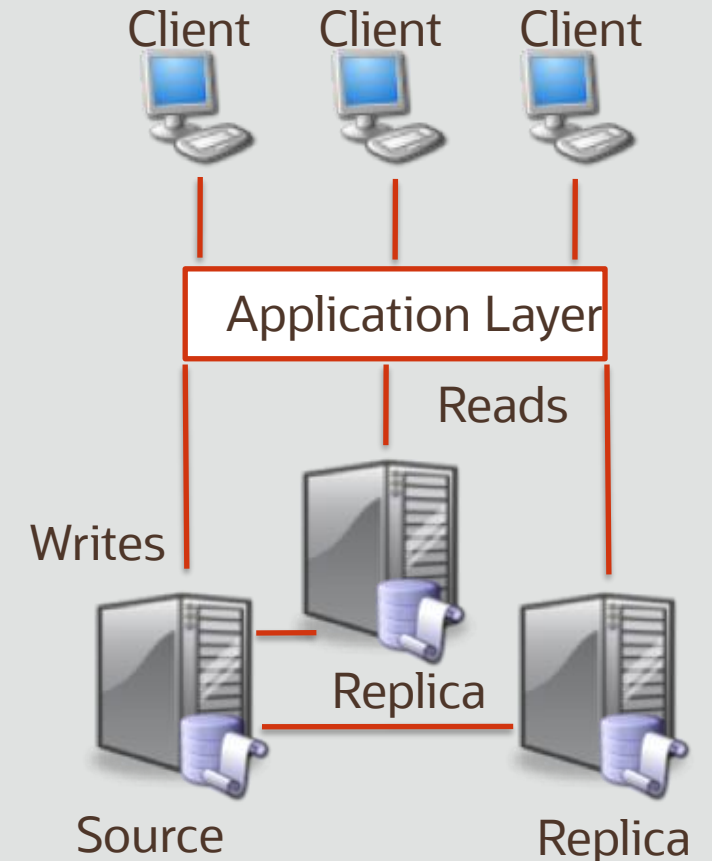


Program Agenda

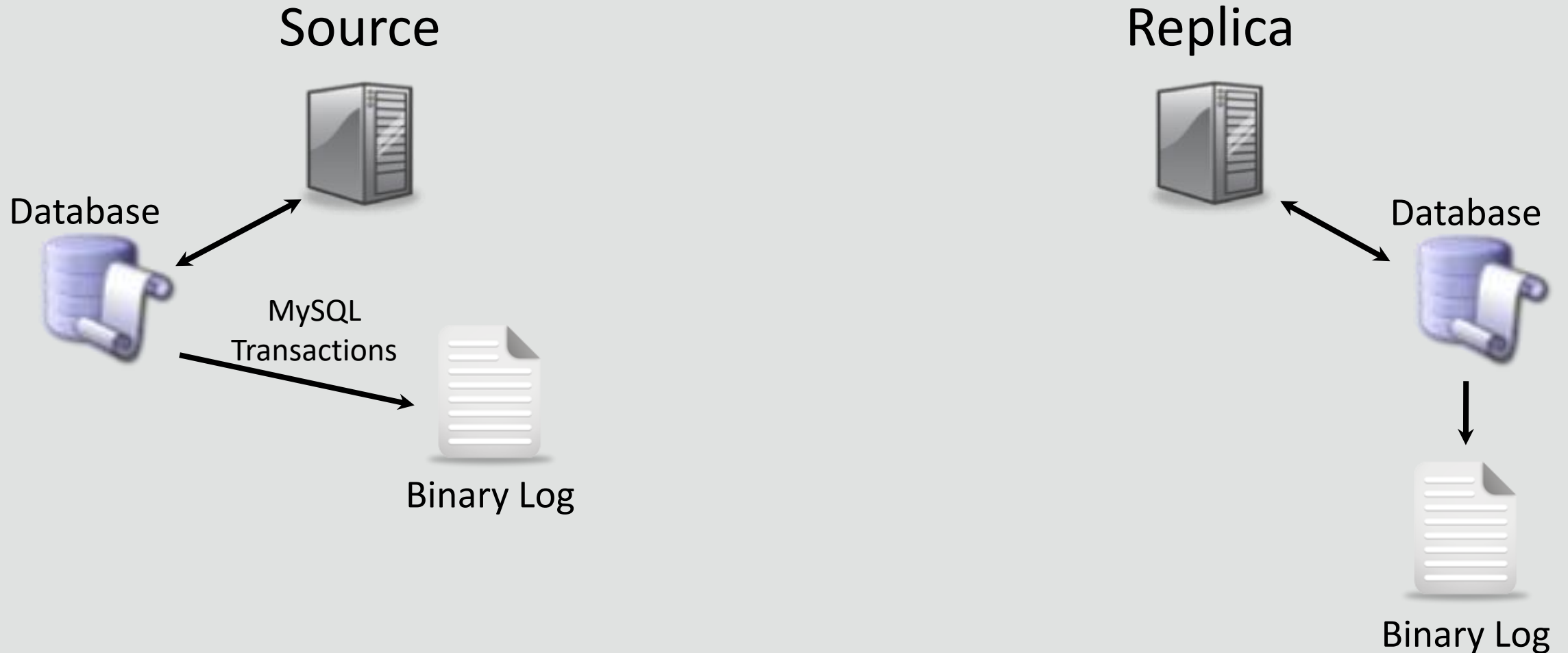
- Introduction
- Oracle MySQL Enterprise
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- Backup
- **Replication**
- High Availability
- MySQL Enterprise Monitor
- HeatWave

MySQL Replication

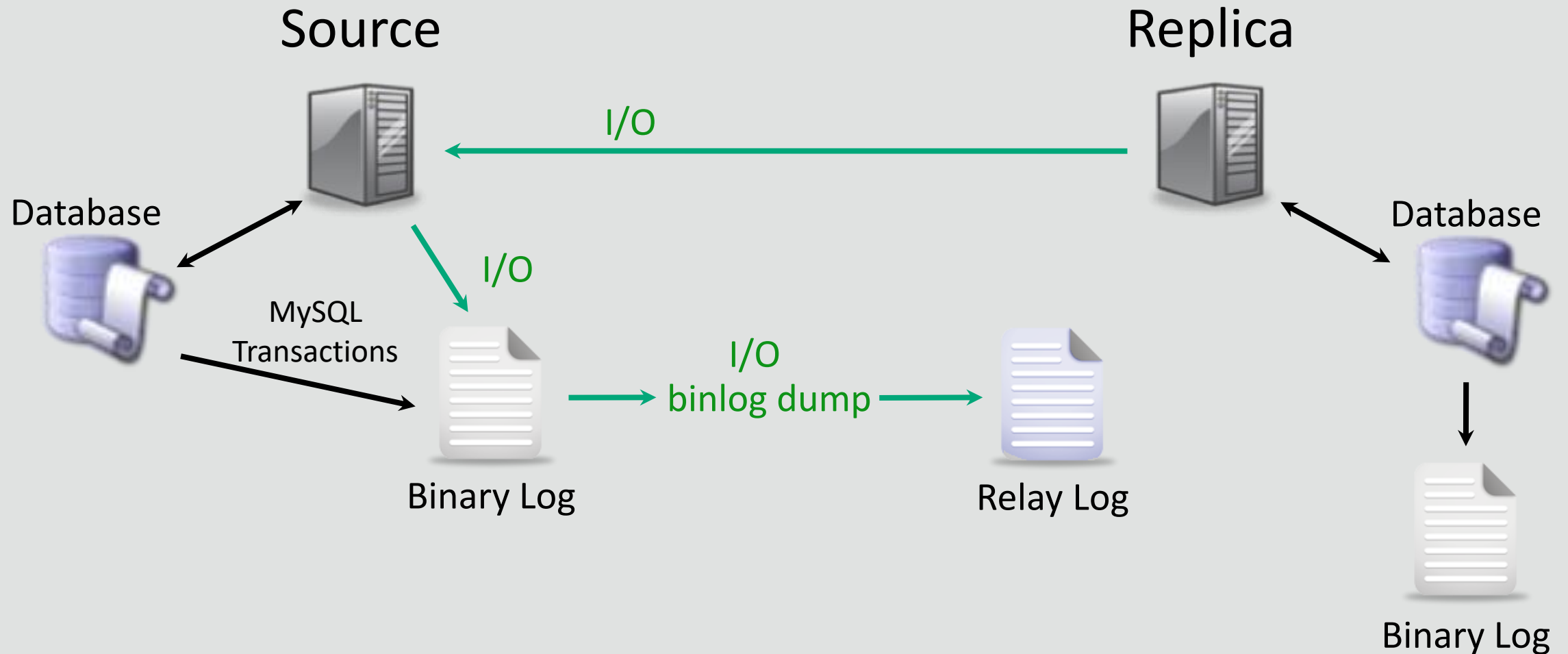
- Data is copied from source to replica
- Replication is asynchronous by default
 - Can be also semi-synchronous
- Multiple source or replicas are possible
- Additional replicas don't require Source configuration
- Target uses for replication:
 - Scale-out solutions for read-heavy environment
 - scaling out doesn't involve editing my.cnf file
 - Simple High Availability (requires scripts and/or external tools)
 - Analytics
 - Long-distance data distribution



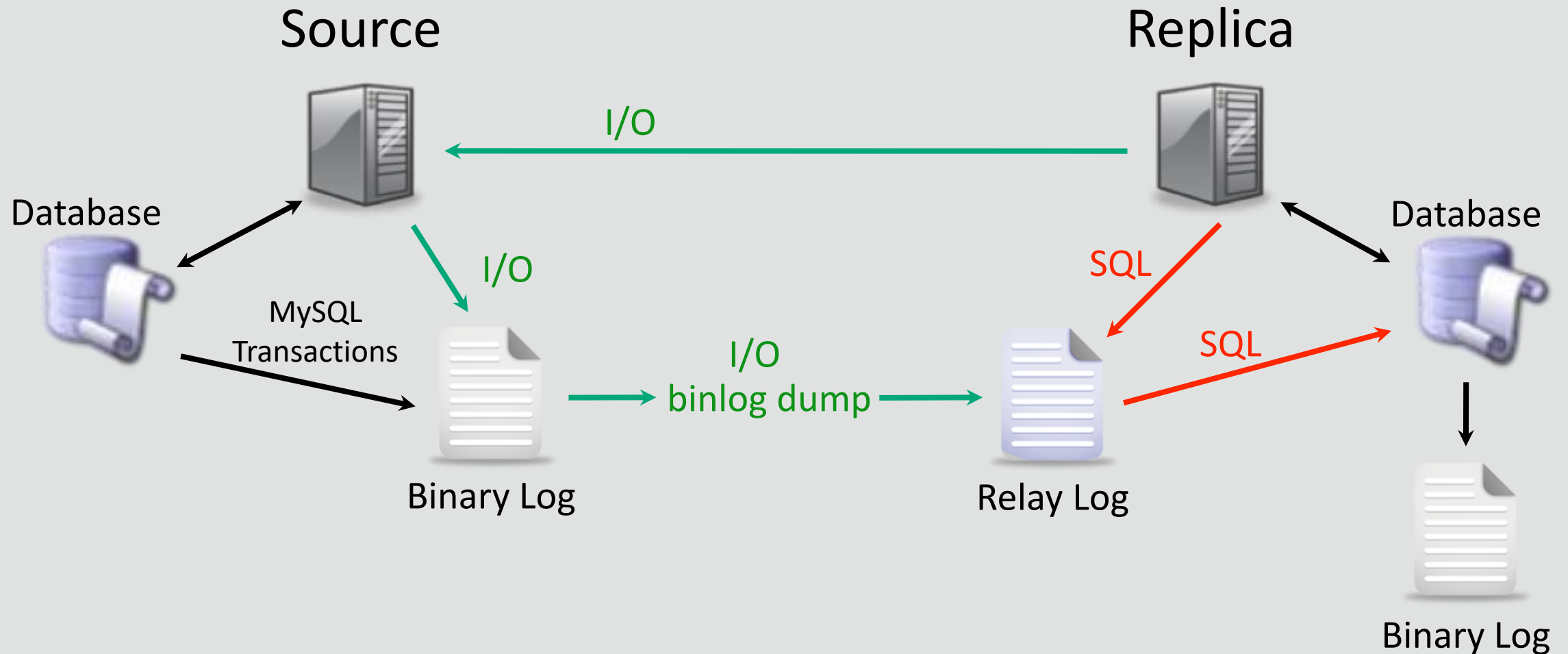
MySQL Asynchronous Replication



MySQL Asynchronous Replication

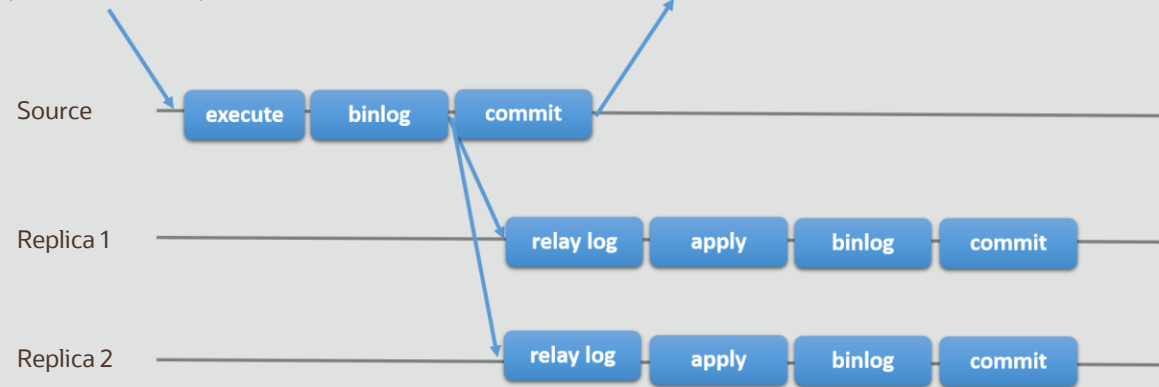


MySQL Asynchronous Replication

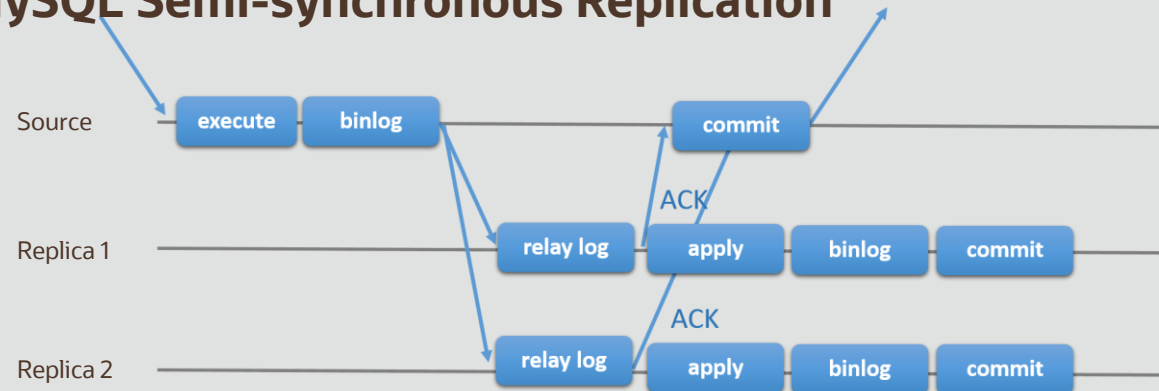


MySQL Replication communication

MySQL Asynchronous Replication



MySQL Semi-synchronous Replication



Replication steps/commands summary

- After both servers have the same dataset
- Create replication user on source
- Check Master configuration
 - SHOW MASTER STATUS
 - Confirm GTID is enabled
- **Configure replica server**
 - CHANGE MASTER/SOURCE TO
- **Start replica**
 - START SLAVE/REPLICA
 - SHOW SLAVE/REPLICA STATUS

Create Replication User on Source

```
mysql> CREATE USER 'replicate'@'%' IDENTIFIED BY 'rep999';
```

```
mysql> GRANT REPLICATION SLAVE ON *.* TO 'replicate'@'%';
```

```
mysql> select user, password, host from mysql.user where user = 'replicate';
```

+	-----+	-----+	-----+
	user	password	host
+	-----+	-----+	-----+
	replicate	*B2063129FC7E0FBED6D0DAB668DF3F75C7AAA851	%
+	-----+	-----+	-----+

```
1 rows in set (0.00 sec)
```

Show Source (Master) Status

```
mysql> show master status\G
```

```
***** 1. row *****
```

```
File: mysql-bin.000005
```

```
Position: 341
```

```
Binlog_Do_DB:
```

```
Binlog_Ignore_DB:
```

```
Executed_Gtid_Set: a5d45538-c22e-11e2-afc1-e4d0658674b1:1-1209
```

```
1 row in set (0.00 sec)
```

Confirm GTID is Enabled

```
mysql> show global variables like '%GTID%';
```

Variable_name	Value
enforce_gtid_consistency	ON
gtid_executed	a5d45538-c22e-11e2-afc1-e4d0658674b1:1-1209
gtid_mode	ON
gtid_owned	
gtid_purged	

```
5 rows in set (0.00 sec)
```

- gtid_owned - session scope: holds all GTIDs that are owned by this client
- gtid_purged -The set of all transactions that have been purged from the binary log

Configure replica server

```
mysql> CHANGE MASTER TO
-> MASTER_HOST = '192.168.1.181',
-> MASTER_PORT = 3306,
-> MASTER_USER = 'replicate',
-> MASTER_PASSWORD = 'rep999',
-> MASTER_AUTO_POSITION = 1;
```

Query OK, 0 rows affected, 2 warnings (0.43 sec)

```
mysql> show warnings\G
```

```
***** 1. row *****
```

Level: Note

Code: 1759

Message: Sending passwords in plain text without SSL/TLS is extremely insecure.

```
***** 2. row *****
```

Level: Note

Code: 1760

Message: Storing MySQL user name or password information in the master.info repository is not secure and is therefore not recommended. Please see the MySQL Manual for more about this issue and possible alternatives.

2 rows in set (0.00 sec)

MySQL 8.0.23+

```
mysql> CHANGE REPLICATION SOURCE TO
-> SOURCE_HOST = '192.168.1.181',
-> SOURCE_PORT = 3306,
-> SOURCE_USER = 'replicate',
-> SOURCE_PASSWORD = 'rep999',
-> SOURCE_AUTO_POSITION = 1;
```

Show Replica Status - Not Running

- When replica is first configured, don't start immediately

```
mysql> show slave status\G
***** 1. row *****
      Slave_IO_State:
          Master_Host: 192.168.1.181
          Master_User: replicate
          Master_Port: 3306
          Connect_Retry: 60
          Master_Log_File:
      Read_Master_Log_Pos: 4
          Relay_Log_File: macos-108-repl02-relay-bin.000001
          Relay_Log_Pos: 4
      Relay_Master_Log_File:
          Slave_IO_Running: No
          Slave_SQL_Running: No
.....
      Executed_Gtid_Set: g39de0f6-c24a-11e2-b077-68dc19e53454:1
.....
```

MySQL 8.0.22+

mysql> SHOW REPLICA STATUS\G

Start Replica

```
mysql> start slave;
Query OK, 0 rows affected (0.01 sec)
mysql> show slave status\G
***** 1. row *****
      Replica_IO_State: Waiting for master to send event
        Source_Host: 192.168.1.181
        Source_User: replicate
        Source_Port: 3306
        Connect_Retry: 60
        Source_Log_File: mysql-bin.000003
  Read_Source_Log_Pos: 803292
        Relay_Log_File: macos-108-repl02-relay-bin.000002
        Relay_Log_Pos: 803502
  Relay_Source_Log_File: mysql-bin.000003
    Replica_IO_Running: Yes
    Replica_SQL_Running: Yes
    .....
```

MySQL 8.0.22+

`mysql> START REPLICA;`

Start Replica (contd.)

```
      Master_UUID: a1a1bcf0-c24a-11e2-b077-adf4bedb11bf
    Master_Info_File: /usr/local/mysql/data/master.info
      SQL_Delay: 0
    SQL_Remaining_Delay: NULL
Slave_SQL_Running_State: Slave has read all relay log; waiting for the slave I/O thread to update it
    Master_Retry_Count: 86400
      Master_Bind:
Last_IO_Error_Timestamp:
Last_SQL_Error_Timestamp:
      Master_SSL_Crl:
    Master_SSL_Crlpath:
Retrieved_Gtid_Set: a1a1bcf0-c24a-11e2-b077-adf4bedb11bf:1-1209
Executed_Gtid_Set: a1a1bcf0-c24a-11e2-b077-adf4bedb11bf:1-1209
      Auto_Position: 1

1 row in set (0.00 sec)
```

Show Replica and Source Status

```
mysql> show replica status\G
***** 1. row *****
....

      Relay_Log_File: macos-108-repl02-relay-bin.000001
      Retrieved_Gtid_Set:
      Executed_Gtid_Set: a1a1bcf0-c24a-11e2-b077-adf4bedb11bf:1-1209
      Auto_Position: 1

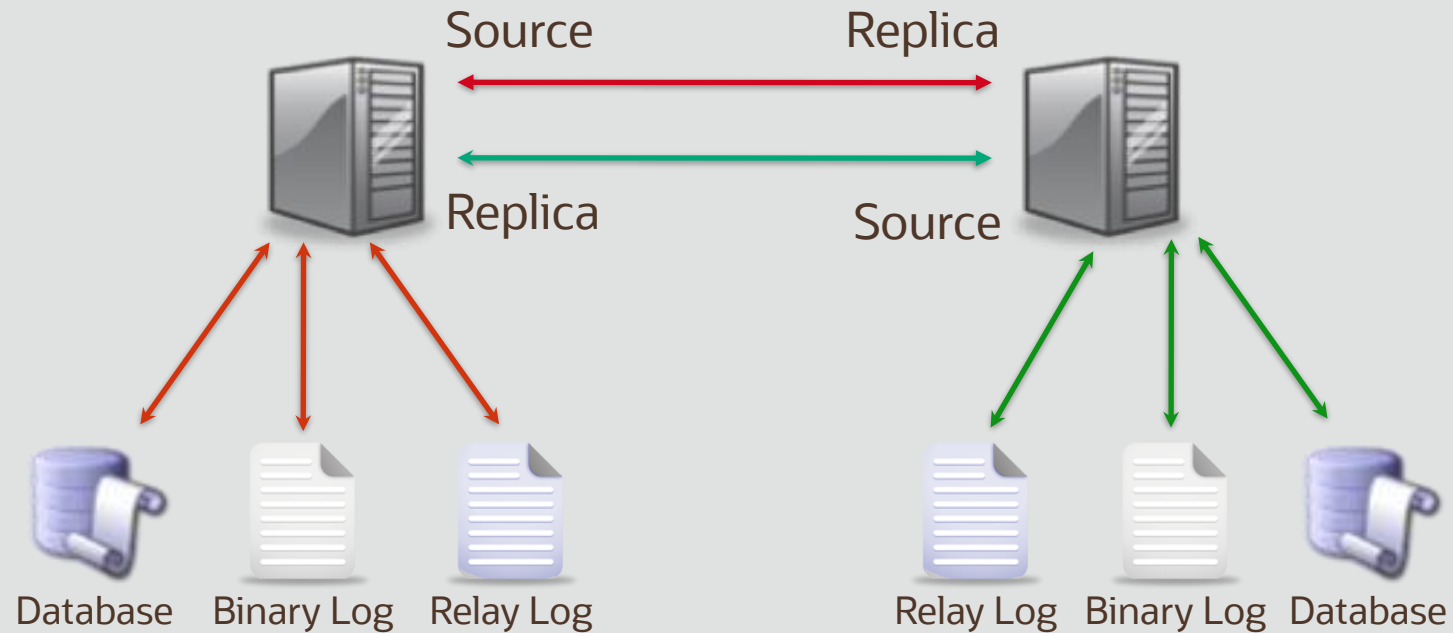
1 row in set (0.00 sec)
```

```
mysql> show master status\G
***** 1. row *****

      File: mysql-bin.000007
      Position: 473
      Executed_Gtid_Set: a1a1bcf0-c24a-11e2-b077-adf4bedb11bf:1-1209

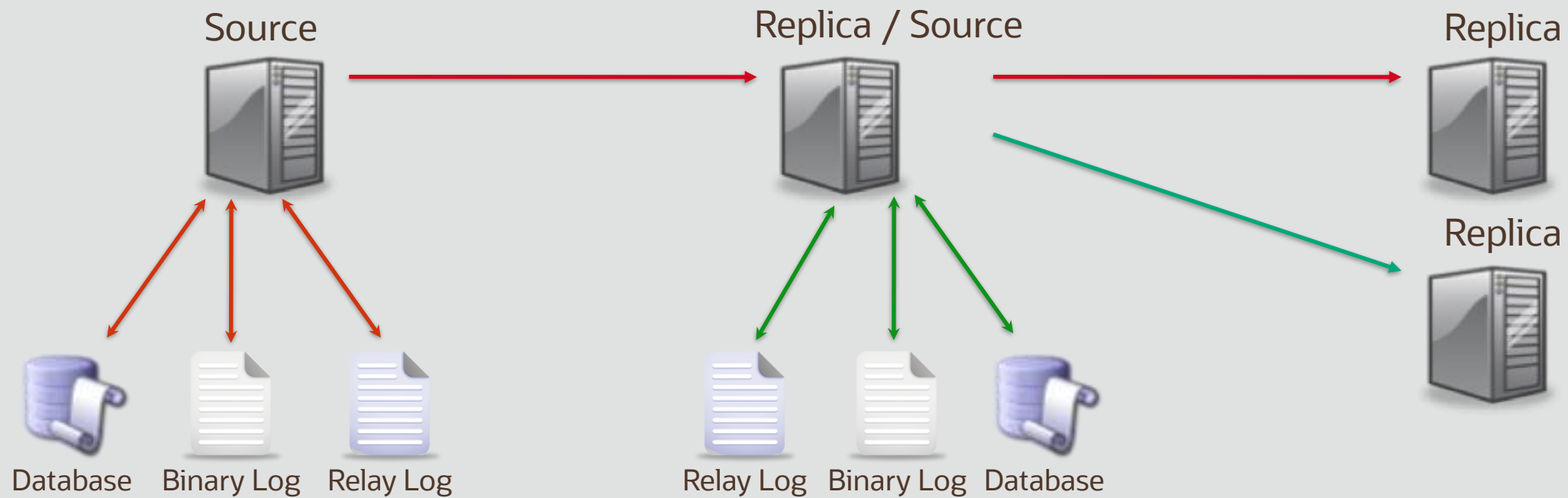
1 row in set (0.00 sec)
```


MySQL Source-Source

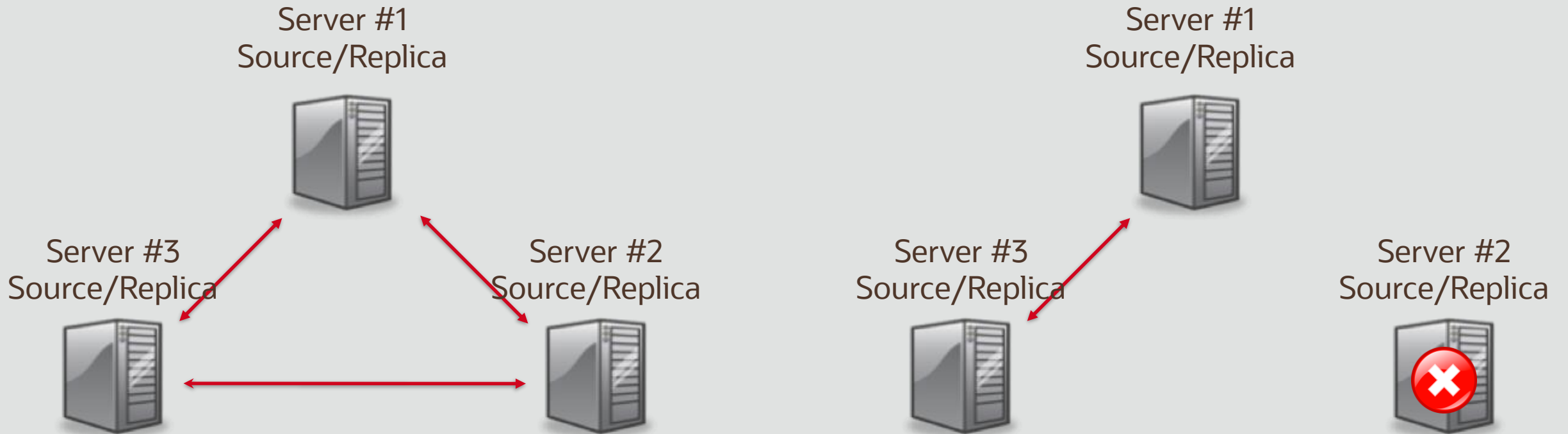


- **No conflict resolution**
 - High risk of corruption due to asynchronous changes
- **Set auto_increment options to reduce risk of conflict, e.g.**
 - FIRST SERVER: `auto_increment_offset=1; auto_increment_increment=2`
 - SECOND SERVER: `auto_increment_offset=2; auto_increment_increment=2`

MySQL Chain

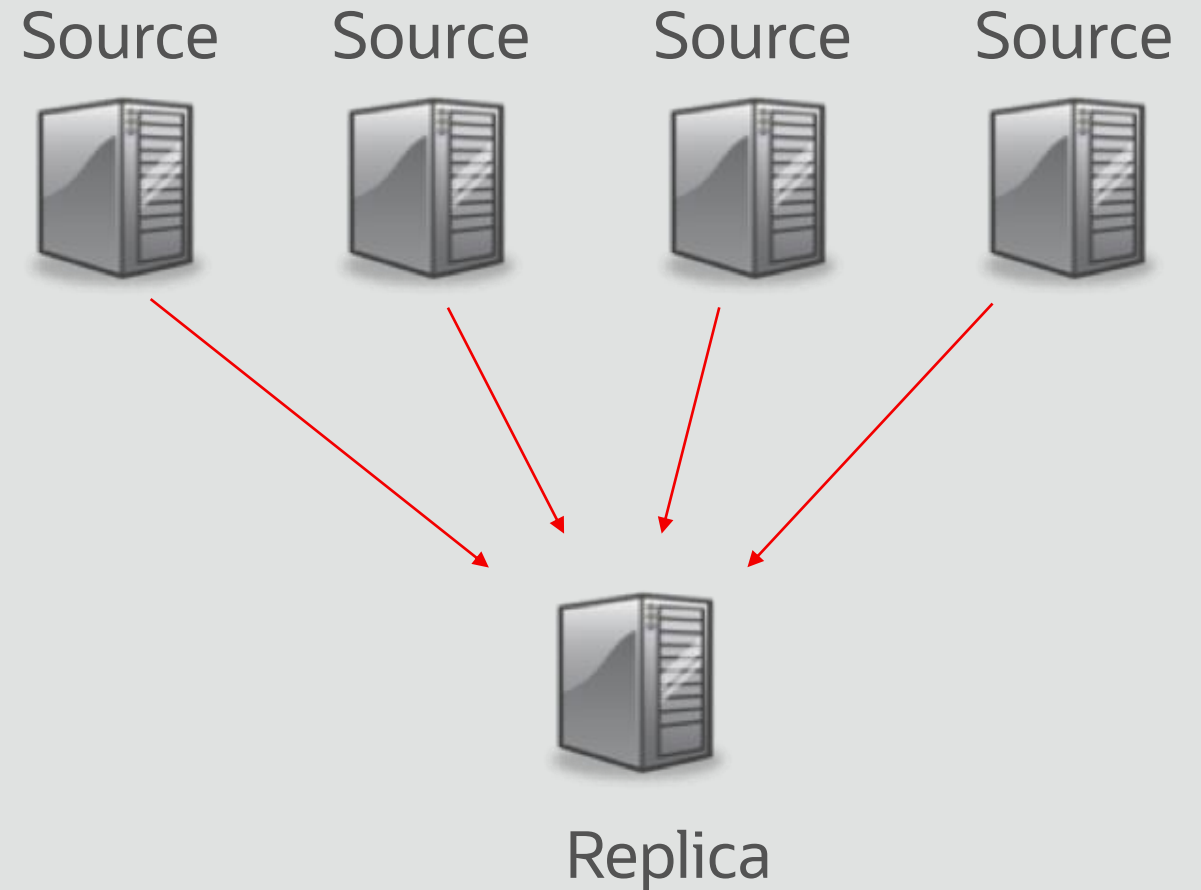


MySQL Replication - Ring-Topology



Multi-Source Replication

- Consolidate updates from multiple sources into one replica
- Compatible with semi-sync replication
- Online operations: dynamic replication filters, switch source





Hands-On Labs



6a. MySQL Replication: prepare Replica server

6b. MySQL Replication: create Replica



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